up with [this idea] at about the same time," and he then leaves the impression that he himself deserves the greater share of the credit but says that "Zel'dovich, Yuri Trutnev, and others undoubtedly made significant contributions."EBM pepst

235 ["In a great number of cases . . . not to grant clearance."] USAEC (1954).

235 [Teller had "had the courage . . . deserved consideration,"] J. A. Wheeler, telephone conversation with K. S. Thorne, July 1991.

235 [Andrei Sakharov, . . . came to agree.] Sakharov (1990).

239 [At Livermore . . . produced a black hole.] The motivation for this research, a quest to understand supernovae and their roles as sources of cosmic rays, is described in Colgate and Johnson (1960). Colgate and White (1963, 1966) carried out the small-mass, supernova-forming simulations, using Newton's description of gravity rather than Einstein's. May and White (1965, 1966) did the large-mass, black-hole–forming simulations, using Einstein's general relativistic description of gravity.

240–241 [To puzzle out the details . . . nearly identical to the Americans'.] Imshennik and Nadezhin (1964), Podurets (1964).

244 ["You cannot appreciate . . . true simultaneously,"] INT-Lifshitz.

244 [Then one day in 1958, . . . David Finkelstein,] Finkelstein (1958).

244 [Footnote 5: See, e.g., the discussions in Box 31.1 and Chapter 31 of MTW.

245 [Finkelstein discovered, quite by chance . . . and stellar implosion.] For Finkelstein's description of how the discovery was made, see Finkelstein (1993).

246 [an article in *Scientific American.*] Thorne (1967).

254 [In 1964 and 1965 . . . stellar implosion.] Harrison, Thorne, Wakano, and Wheeler (1965).

256 [He tried it out at a conference . . . "By reason . . . increase its gravitational attraction."] Wheeler (1968).

CHAPTER 7

258 General comment about Chapter 7: The historical aspects of this chapter are based on (i) my own personal experience as a participant, (ii) my interviews with other participants (INT-Carter, INT-Chandrasekhar, INT-Detweiler, INT-Eardley, INT-Ellis, INT-Geroch, INT-Ginzburg, INT-Hartle, INT-Ipser, INT-Israel, INT-Misner, INT-Novikov, INT-Penrose, INT-Press, INT-Price, INT-Rees, INT-Sciama, INT-Smarr, INT-Teukolsky, INT-Wald, INT-Wheeler, INT-Zel'dovich), and (iii) my reading of the scientific papers the participants wrote.

262 ["There have been few occasions . . . consummated."] Wheeler (1964b).

266 *[hoop conjecture:*] I first published the concept of the hoop conjecture in a Festschrift volume in honor of Wheeler (Thorne, 1972), and in Box 32.3 of MTW.

268 [the idea that the implosion of a star . . . run backward.] This idea was called by Novikov and Zel'dovich the *semiclosed universe.* They ultimately published separate papers describing it: Zel'dovich (1962) and Novikov (1963).

269 ["Maybe you *now* don't . . . but you *will* want to."] INT-Novikov.

269 ["Yakov Boris'ch would . . . next day."] INT-Novikov.

274 [To test this speculation . . . no magnetic field whatsoever.] The key ideas and initial calculations of this research were published in Ginzburg (1964); more complete